To	Group Director, Credit Risk
From	Retail Banking
Date	October 2023
Subject	**REQUEST FOR PEZESHA SCORECARD APPROVAL**

Part 1: Background

In line with growth objective of acquiring one million new customers and 67% loan book growth, Retail Digital banking, intends to introduce short term, unsecured Digital loans which will be managed separately and outside the conventional NCBA systems. The solution targets Small Medium and Enterprises with DFS as the anchor product, direct SME lending and supply chain Finance.

The database, Loan management system and mobile application will be housed within NCBA. The product will be offered in partnership with Pezesha who will provide e2e white-labelled Loan origination and loan management system with an end of day entry into the NCBA general ledger (GL) for reporting purposes as well as integrate with third party data aggregators including Safaricom, credit reference bureau(s) and IPRS.

The loans will be accessed through a white-labelled USSD, embedded Finance API, mobile and web application.

Pezesha is a Kenya based Fintech, approved by CBK, processing more than Ksh 350M loans monthly. Pezesha's customizable infrastructure has been operating for the past 6 years and can be applied vertically and horizontally across value chains. They are enabling digital lending to MSMEs through Embedded Finance technology services to FinTech distributors like Twiga Foods, Market force, Jumia and direct SME lending on their platform.

a) Problem Statement- What problem are we solving for?

Below are the key customer pain points that the proposition intends to solve for:

- Manual Onboarding process: - No current capability of Digital KYC and customer on-boarding.
- Lengthy credit appraisal process: - where security is available, the associated timelines for appraisal and credit decision is lengthy resulting to lost business opportunity.
- Data: - Unavailability of data to underwrite SMEs for financing and where available, the data is not structured for credit decisioning
- Lack of bespoke solutions to meet emerging business and industry needs.

b) Value Proposition- What do we intend to achieve?

The proposed partnership is based on business value that the bank intends to unlock. Some of the key value drivers include:

- New Customer acquisition and growth: - using an e2e Loan Origination System (LOS) and loan Management System (LMS)

- Balance sheet growth: - Existing customer base and loan book to be scaled up with other cross sell opportunities
- Data: - Access to rich data of Banking ready customers from various embedded partners to be monetized.
- Technology: - White-labelled solution offering Digital KYC, customer onboarding, loan origination and loan management system.
- Improved customer experience: - due to seamless processes and reduced TAT
- Cost Reduction: - due to automated processes and operational efficiency.

Project scope

The project scope includes:

Phase 1: **Embedded Finance - Customer acquisition and lending**

- The scope for phase 1 will primarily focus on customer acquisition and loan book growth through digital lending. Below is what is covered within the scope of phase 1.

EKYC and Digital Onboarding

- There will be tool for eKYC and Digital onboarding to assist with NTB customer acquisition. Channel of access for customers will be through the App.

Loan Origination System (LOS)

- The loan origination system will have three channels of access for customers: App, USSD and Web service. Customers will choose their channel of choice.

Loan Management System (LMS)

- The loan management system will handle credit scoring, limit assessment, customer profile and limit management.

Collection

- Collection for 1-30 DPD will be handled by Pezesha to assist improve on portfolio quality.
- Embedded Finance API which will enable the bank have access to rich ecosystems of partners with existing customers and data that can be monetized.

Implementation plan

- Project will be piloted with the current DFS customer base for a period of 6 months with the intention of evaluating and enhancing the scorecard.

Phase 2: Direct SME & Cash management services

The primary objective of the project/initiative is customer acquisition and loan book growth. Once the customers are acquired to bank, we intend to offer additional banking services starting with Direct SME onboarding & lending plus cash management services, including:

- Payments
- Collections
- Account services, among others
- This will help the bank significantly grow its customer base, liability and asset book and revenue.

The partnership model is on revenue share. However, below is why the partnership is unique, compared to our existing partnerships:

- **Cost/Capex:** - The is no capex or initial implementation cost to the bank in this partnership.
- **Risk sharing model**: Revenue share model based on loan performance. The partner will unlock their share of revenue based on the loan book performance.
- **Collection and portfolio management**: - Support on collection and portfolio management for 1-30 DPD cohort.
- **Access to alternative data** through different ecosystems and/or partners to help in customer acquisition and loan book growth.

a) <u>Embedded Finance Product Features</u>

Loan amount	Minimum loan amount: from KES 500 Maximum loan amount: KES 500,000
Security	▪ Unsecured
Tenor	▪ Based on nature of product and product conversion cycle ▪ Minimum tenor: 7 days ▪ Maximum tenor: 90 days
Loan price	▪ Facility fee: Up-to 4% ▪ Interest: Base rate + 4.1% (pro-rated for the period of the loan)
Disbursement options	▪ Directly to **supplier's (Anchor) NCBA account** with appropriate narrations ▪ Disbursement to M-pesa till/paybill
Channel of access	▪ Mobile App and USSD
Method of Repayment	▪ M-Pesa stk push ▪ NCBA account through mobile USSD and App ▪ Automatic debit through T24 COB or sniffing tool

<u>Scorecard Details</u>:
Credit scoring is a statistical method used to predict the probability that a loan applicant, existing borrower, or counterparty will default or become delinquent.
A **credit scorecard** consists of a **group of borrower characteristics**, statistically determined to be predictive in **separating good and bad loans/customers**. Each attribute is assigned **points** based on statistical analyses. The sum of the individual scores of the characteristics provides an overall **credit score**, which reflects the overall likelihood a customer will be a good/bad borrower.

<u>Methodology</u>
There a number of options for credit scoring (regression, logistic scorecards, machine learning, rule based etc.). The preferred (and industry standard) is the **logistic regression and weight of evidence approach**.
Pros:

1. **Interpretability** - easy to understand why a customer has been given a certain score - can easily explain to stakeholders, customers or regulators.
2. **Transformation -** of an independent variable so that it establishes **monotonic relationship** to the dependent variable. the WOE approach transforms the variable onto a "logistic" scale.
3. **Data quality/missing values** - for variables with many (sparsely populated) discrete values, these can be grouped into categories (densely populated) and the WoE can be used to express information for the whole category. It also is very useful for handling outliers a can leverage the information value of **missing data**.
4. **Data** - This approach doesn't require significant amounts of data and is less prone to overfitting than many machine learning methods.

Cons:

1. Potential **loss of information** due to binning too few categories - this can me minimised by robust analysis and testing.

2. Does not account for the **correlation** between categories (univariate).
3. Can lead to **overfitting** if too few/many categories are chosen.

The purpose of the **Weight of Evidence** (WoE) is to assign a unique value to each group of borrowed characteristics, which reflects that groups good/bad. The **Information Value** (IV) measures predictive power of the characteristic, which is used for feature selection.
The definition of **'bad'** in this case is a loan that has been paid late by > 3 times the duration of the loan.

Pezesha uses the **Weight of Evidence** (WoE), which measures the predictive power of customer characteristics by taking the natural logarithm of the proportion of good loans to bad loans for each category. The **Information Value** (IV) measures strength of the characteristic, which is used for feature selection.
The bins are chosen to **maximize the distance between good/bad** customers as a monotonic increasing/decreasing function. Characteristics are then chosen with the highest IV, e.g. < 0.1 = weak, 0.1 - 0.3 = good, > 0.3 = strong. WOE values are then mapped back to the characteristics in the dataset and borrowers/loans are put into their respective bins before being passed through a **logistic regression model**.

Scoring Model and Borrower Eligibility Criteria

Embedded Finance
All merchants are subject to the following minimum requirements: 1. Active in the last 30 days 2. Minimum 3-months history with the anchor partner 3. Transacted with the anchor partner in the last 15 days (sometimes more depending on the business model, e.g. healthcare, logistics or tour operators have longer product cycles). 4. Full KYC data (national id, name, phone number, gender, age, location, KRA pin). 5. Minimum number of transactions to model (usually > 20). 6. No outstanding loan with Pezesha and no non-performing accounts.
In addition, embedded finance merchants must pass the following pre-scoring checks: average weekly revenue >=2000 ksh active days > 90 Days since last transaction <=15 Non-performing accounts <=1

Scoring Data and Model Characteristics

Embedded Finance
Know Your Customer (KYC) Location, Business Age, Business Size, Sector
Transactional data a. Recency, Frequency, Monetary (RFM) b. Trend c. Spending d. Transaction stability e. Longevity f. Distribution (centrality/dispersion) g. Churn

Embedded Finance Minimum Data Requirements

a) KYC Data

Merchant_id	Unique merchant identifier in partner system
National_id	Unique identification number issued to Kenyans that are 18 years and above or Unique passport Number
PIN	Personal Identification Number (PIN) as issued by Kenya Revenue Authority (KRA)
Nationality	Customer's Nationality based on the ISO country code of the customer (party)'s Nationality.
Merchant_Name	Full name of merchant
Merchant_County	County the business is registered in
Merchant_Sub-County	Sub-County the business is registered in
Merchant_reg_date	Date Merchant was first registered in Pezesha partner's system
Merchant_DOB	Date of Birth (**Y-m-d format**)
Mobile_number	Mobile Phone Number (CCC-AAA-XXXXXXX: CCC -Country Code, AAA - Area Code and XXXXXXX - Contact Number).
Merchant_gender	M/F
geo_location.long	

b) Transactional Data

Item #	Column Name	Detailed definition of column elements	Data Type	Mandatory Fields
1	ROW ID	Sequential number starting from the first number for the first record.	Number	Y
2	CHANNEL	Unique partner name	Character	Y
3	MERCHANT_ID	The unique merchant identification number	Characters/Number	Y

4	TRANSACTION_ID	The unique transaction identification number, associated to the merchant transaction	Character/Number	Y
5	FACE_AMOUNT	The face amount of the transaction, reported in Kenyan Shillings.	Number	Y
6	TRANSACTION_DATE	The date the transaction took place. Using the format {**YYYY-MM-DD**}.	Date	Y

Scorecard Weights- Embedded Finance

Credit scoring modules can be combined into a single scorecard OR they can be combined/aggregated by weighting. Pezesha's current weighting structure for embedded finance is below (this depends on the anchor partner):

Embedded finance	**New/repeat**	**weight**
Transaction data / kyc	new	0.8
Loan data	new	0
Macro-economic data	new	0.2
Transaction data / kyc	repeat	0.2
Loan data	repeat	0.7
Macro-economic data	repeat	0.1

Embedded Finance Credit limits

Credit limits are a function of borrower credit scores and transaction values. Repeat customer can graduate their loan limit subject to their credit performance. At every loan application – the minimum criteria are re-checked as well as credit history (No prior NPA, delinquency or defaults)

New customer	**Repeat customer**
Low Risk (score >210): 40% of average monthly transactions	10% increase of previous loan offer
Medium risk: (180 < score <210): 30% of average monthly transactions	5% increase of previous loan offer
High risk : (Score >150 < 180): 20% of average monthly transactions	0% increase of previous loan offer

Credit Administration:

Portfolio Management

SMS notifications will be sent promptly and will be categorized as below:

- Transaction Notifications:

This will be sent on real-time basis and will relate to below transactions:

- Loan limit check
- Loan disbursement
- Loan balance check
- Loan repayment
- Portfolio Management messages

This category of messages will be sent based on the age of the loan. There will be loan ageing service that will assist manage the loan schedule and trigger release of the messages at the appropriate time. The messages will be sent before, on and after due date.

Portfolio monitoring and Risk triggers

Key Triggers	**Target**	**Portfolio Triggers**
Frequency of late repayment (>15 days)	Once a month for three consecutive months	Aggressive early collection efforts by RED team
Persistent late repayments (>30days) for more than 5% (by number) of distributors of any particular supplier under the scheme.	6.5%	RM/Product manager to engage the partner/supplier to review any challenges and support on customer engagements
NPL @ portfolio/scheme level	5%	a) Heighten collection and recovery efforts b) Invoke partnership clauses on: • Stop orders • Support on collection
Any material change/decline in the number of retailers/distributors by the Supplier	50%	a) Close portfolio monitoring b) Suspend lending c) Deploy exit strategy
Any event that affects the supplier's market presence that might adversely change the marketability of its products.	To be monitored	a) Suspend lending b) Deploy exit strategy

Collection process

1. **Digital Loans**

<table>
<tr><th>Activity</th><th>Role</th><th>Action</th></tr>
<tr><td>Automated Notifications</td><td>Products, Channels & Digital Banking via Sasa Platform

PEZESHA PLATFORM/CHANNELS</td><td>1. Automated SMS from SASA is sent to the customer upon disbursement of the loan facility, indicating amount disbursed and the due date.
2. A prompt SMS is automatically sent to the customer via the SASA platform to remind him or her of the due date on the following days:
<table><tr><th>One Month Loan</th><th>One Week Loan</th></tr><tr><td>29th day</td><td>6th day</td></tr></table>PEZESHA LOANS
1. An automated message is sent to each borrower 1 day upon disbursement indicating amount disbursed and the due date.
2. 5 days before the whole loan or installment loan is due, a prompt SMS is automatically sent to the customer via the SASA platform to remind him or her of the due date
3. 3 days and 1 day before the loan is due a PEZESHA Collection Officers call the borrower to remind him /her for payment.
● Collection Officers will work on a process defined by the Collections Manager</td></tr>
<tr><td>Collection
1 dpd-7 dpd</td><td>1. Retail early collections (R.E.D)

PEZESHA PLATFORM/CHANNELS
1 DPD -30 DPD</td><td>2. Accounts over 0 days in arrears are queued automatically in the collection system i.e. Kollect Pro.
3. Kollect Pro dispatches sms demand and notifications, twice per day (in the morning and in the evening)
4. Kollect Pro dispatches email notifications and demand twice per day (in the morning and in the evening)

PEZESHA LOANS
LATE MERCHANT WORK
The touchpoints for late merchants is as below
• SMS: 5 SMSs
• Phone call: 5 calls
• Field visit: 2 field visits
• Internally generated Demand notice- 2 notices
• Demand letter - 2 letters</td></tr>
<tr><td>Collection
8 dpd - 90 dpd</td><td>Remedial Officer</td><td>1. Daily extraction of reports off Kollect Pro on the dispatched demand SMS and \or emails sent previous day and send to the</td></tr>
</table>

		Products and Digital Banking team for purposes of monitoring the trend in volumes\values falling into arrears daily and monitoring the performance of the automated messaging service on Kollect Pro. 2. Follow up and review of the daily reports from: ➢ Collect Pro ➢ Collection Agencies ➢ Core Banking 3. Daily restriction of accounts with overdue balances and submission of a report on the restricted account to Products and Digital Banking team.
Collection 8 dpd - 90 dpd	Business Remedial Management/External Collection Agencies	1. The Collection System-Kollect Pro, will automatically allocate accounts over 8 DPD to external collection agencies. • PEZESHA LOANS: • The Collection System-Kollect Pro, will automatically allocate PEZESHA loan accounts over 30 DPD to external collection agencies. 2. A formal notification to the customer from the Bank through Kollect Pro will be sent informing them that their debts have been outsourced to a collection agency at their cost. This will be done through a letter/email and notification SMS. 3. The external debt collection agency will follow up with the customer through SMS/calls/Emails and field visits. 4. The amount outsourced will include the total outstanding amount with the Bank plus the related collection charges by the debt collector. 5. The collection charges levied will be based on a graduated scale negotiated with the collection agencies. 6. Payments collected will be banked to the respective customers' accounts with us. 7. Payments will then be applied to the respective loan accounts and payoff done. 8. The external collection agencies will then raise their invoices at the end of every month for review and settlement. 9. Invoices will be settled within 14 days of receipt by the Bank. 10. Reallocation of accounts between collection agencies will take place after

		30 days for both the 1-month loan and the 1-week loan. 11. Reports from the agencies will be provided to BRM and the Product & Digital Banking team every 30 days from the date of allocation of a batch of accounts
Performance	Portfolio/Recoveries Officer Live and B20	12. A weekly performance report will be shared with the Products, Digital Banking and BRM team to show performance of the book and recoveries from ICAs shared
CRB Pre and Post-Listing 30 dpd-90 dpd	Credit Operations Officer	13. Send letters\email and SMS notifying the customers of pre and post-listing status as per CBK guidelines.
Over 90 dpd	Business Remedial Management	14. Upon reaching 90 days past due, the loans that will still be outstanding will be recalled from the external debt collection agencies and migrated to Memo/B20 book where accrual of interest will be stopped. 15. Follow up with the customers will continue until they make full payments or accounts are written off.

Risk Grouping

New customers with scores between 275 and 199 will be awarded credit limits. As they transition into repeat customers we will adopt the bank risk rating matrix as per the table below.

		SCORE	
RISK CLASSIFICATION		>	<=
Low risk	A1	275	300
	A2	250	274
Medium risk	A3	230	249
	A4	200	229
	A5	180	199
High risk	A6	150	179
Very high risk (non-fundable)	A7	0	149

Loan grading and risk classification

Loan grading and risk classification will be done as per the CBK prudential guidelines and bank internal credit risk policy:

Grade	Classification	No of days past due
1	Normal	0 – 30
2	Watch	30 – 90
3	Sub-Standard	90 – 180
4	Doubtful	180 - 360
5	Loss	360+

Penalty on default:
The loan to incur a penalty of 2.5% monthly based on the outstanding principal amount.

Loan Loss Provisions
Non- Performing loans will be provided for as per the CBK prudential guidelines and IFRS 9 guidelines. Write off shall follow the laid down process

Reporting Requirements:

a) Reports will be generated from BI and from the portal
b) Customer creation reports –success and failures (status) with failure reasons will be availed on the portal
c) Loan requests –Success and failure (status) with failure reasons
d) Loan repayments - Success and failure (status) with failure reasons
e) Overdue Loan reports, Ageing in days, value and accrued interest and penalty
f) Loans rolled over report
g) Commissions collected report, per day, Month, year and split between the Interest collected and arrangement fees

Expected benefits (ROI)
The expected benefits from this partnership include:

a) Access to potential 200K base of customers and loan book of $ 10,000,000

Pezesha is a going concern Fintech with experience in the SME space. The partnership will be premised on using our balance sheet and leverage on their existing customer base and partners at inception. The intention is to then scale distributor Finance Base Anchor in the same model.

The outlook at 50% penetration of the book is highlighted below:

	Year 1	Year 2	Year 3
Disbursements	9,205,976,436	20,240,162,852	25,183,751,737
Revenues			
Appraisal fees	368,239,057	809,606,514	1,007,350,069
Interest rate	138,089,647	303,602,443	377,756,276
Total	**506,328,704**	**1,113,208,957**	**1,385,106,346**
Costs			
FTP Charge	57,671,786	126,614,775	157,432,946
Business Development People Costs	1,750,000	3,700,000	4,500,000
Marketing Cost	1,200,000	1,200,000	1,200,000
Revenue share to PARTNER	110,471,717	242,881,954	302,205,021
Implementation cost	7,500,000		
Provisions	184,119,529	404,803,257	349,434,905
Total OPEX	**362,713,032**	**779,199,986**	**814,772,872**
Net Income	**151,115,672**	**334,008,971**	**570,333,474**

Revenue Share structure

The revenue share model is based on the performance and is pegged on the Portfolio at Risk (PAR) target net of the cost of capital and provisions

PAR is defined to be the percentage of principal disbursed that is >= 7 days late. PAR is calculated as the outstanding principal value of a portfolio that is late by 7 days divided by the principal disbursed of the entire portfolio.
The

Revenue share	Target
30% Rev share	PAR at less than 10%
20% Rev share	PAR between 10.1- 20%
15% Rev share	PAR between 15.1%- 20%
10% Rev share	PAR between 20.1-30%
5% Rev share	PAR between PAR at more 30.1%

This is the simulation of **best-case scenario** and portfolio is performing at **PAR of less than 10%**

	Year 1	Year 2	year 3
Disbursement	9,205,976,436	20,240,162,852	25,183,751,737
NCBA Share (Gross)	506,328,704	1,113,208,957	1,385,106,346
NCBA share (Net off costs – FTP, provisions etc)	151,115,672	334,008,971.00	416,093,344
Partner share (on fees alone)	110,471,717	242,881,954	302,205,021

Recommendation

Rigorous testing will be done on the scorecard before deployment to determine its efficacy. In addition, regularly reviews will be done to improve the model and make it work better. We recommend approval to enable us proceed to implementation.

Recommendations and approvals:

Proposed by:

Patience Wanjala Senior Manager, Digital Channels	**Supported**

Reviewed by:

Kenneth G. Kimathi Head, Credit Quality Assurance & Compliance	**Supported**

Supported by:

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Edwin Induli Director, Credit Risk Management	**Supported**
Pauline Ndote Group Director, Credit Risk Management	**Supported**
Julius Kamau **Group Director, Technology & Operations**	**Supported**

Appendix



Embedded Flow_Pezesha.pdf